

Sensitive volumes in the G4VM geometries

A volume is read out when its GDML <volume> carries the tag

```
<auxiliary auxtype="SD" auxvalue="name" auxunit="id"/>
```

DetectorConstruction::ConstructSDandField() scans every logical volume for it and attaches one `sensi` instance per distinct *name*. That name is the hit-collection name and appears in the `det` column of the output; volumes sharing a name share one detector.

KET — KET/ket.gdml, 17 sensitive volumes

SD name	Volume	Material	Subsystem file
<i>Cherenkov</i>			Cerenkov/cerenkov_structure.xml
C200	CRC2_V	PBF2	lead fluoride radiator
Aero	CRC1_V	Aerogel	aerogel radiator
Quartz	CC2P_V	Quartz	quartz radiator
Glas	Glass_V	Quartz	window / light guide
<i>Scintillators</i>			Scintillators/scintillators_structure.xml
S2_box	S2box_V	NE104	
S2_SCSB	SCSB_V	NE104	
AC	SCAD_V	NE104	
AC1	SCA1_V	NE104	
AC2	SCA2_V	NE104	
AC4	SCA4_V	NE104	
AC5	SCA5_V	NE104	
AC6	SCA6_V	NE104	
AC7	SCA7_V	NE104	
AC8	SCA8_V	NE104	
<i>Solid-state detectors</i>			SSDs/ssd_structure.xml
SD1	SID1_V	Silicon	
SD1A	SD1A_V	Silicon	
SD2	SID2_V	Silicon	
SD2A	SD2A_V	Silicon	

HET — HET/HET.gdml, 11 sensitive volumes

Identical tags in HET/HET_shield.gdml. Names are numeric.

SD name	Volume	Material	Position in the stack
1	HETAinner1	Silicon	A, inner segment
2	HETAouter1	Silicon	A, outer segment
3	HETBinner1	Silicon	B, inner segment
4	HETBmiddle1	Silicon	B, middle segment
5	HETBouter1	Silicon	B, outer segment
6	Crystal_V	BGO	calorimeter crystal
7	HETBinner2	Silicon	B, inner segment
8	HETBmiddle2	Silicon	B, middle segment
9	HETBouter2	Silicon	B, outer segment
10	HETAinner2	Silicon	A, inner segment
11	HETAouter2	Silicon	A, outer segment

Note. The two silicon groups (1–5 and 7–11) are the mirrored halves of the telescope on either side of the BGO crystal, which is why the trailing 1/2 in the volume names distinguishes the halves.

EPHIN — EPHIN/ephin.gdml, 17 sensitive volumes

The same tags apply to `ephin_soho_shield.gdml` and `ephin_chandra_shield.gdml`: all three include the same `detector/` and `anticoincidence/` structure files. Names are numeric.

SD name	Volume	Material	Source file
10–15	vd10–vd15	Semiconductor	detector/detector_structure.xml
20–25	vd20–vd25	Semiconductor	detector/detector_structure.xml
30	vd30	Semiconductor	detector/detector_structure.xml
40	vd40	Semiconductor	detector/detector_structure.xml
50	vd50	Semiconductor	detector/detector_structure.xml
60	vd60	Semiconductor	detector/detector_structure.xml
100	vAnc	Scintillator	anticoincidence/anticoincidence_structure.xml

The two blocks of six are the segments of the two segmented detectors; the four single volumes and the `anticoincidence` follow. The numbering depicts the usual EPHIN labelling (A, B, C, D, E, F and AC), but the GDML itself only carries the numbers.